

WASP-12b

R-0264

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_241105 · created 2026-07-17T09:29:00+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460619.8316 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1024 +/- 0.0061
Transit depth (Rp/Rs)^2	1.05 +/- 0.13 [%]
Orbital Inclination (inc)	85.65 +/- 1.43
Airmass coefficient 1 (a1)	1.0207 +/- 0.0068
Airmass coefficient 2 (a2)	-0.0194 +/- 0.0027
Scatter in the residuals of the lightcurve fit is	0.66 %
Best Comparison Star	#1 - [326, 228]
Optimal Method	PSF photometry
Transit Duration (day)	0.126 +/- 0.0019

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.1024 ± 0.0061 vs 0.1178 (deviation 1.83σ)
Duration fit vs published	0.126 d vs 0.125 d (deviation 0.38σ)
Mid-time O–C	-3.6 min
Depth SNR	12.1
Residual scatter	0.66 %

Light curve

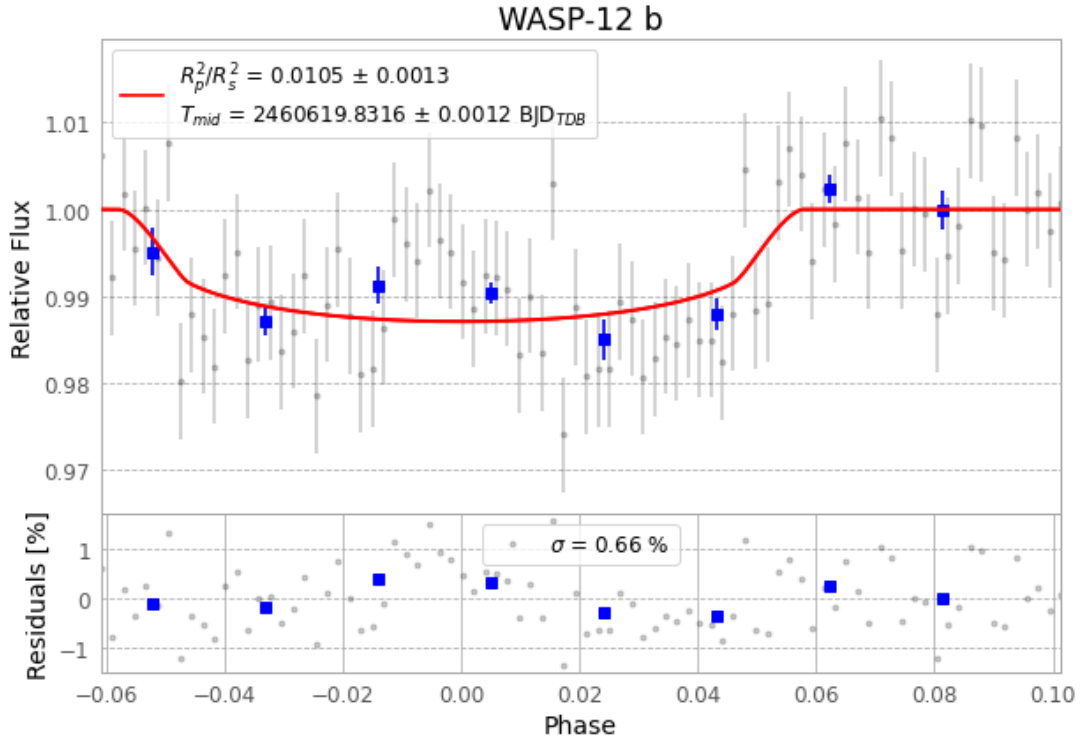


Figure 1 — FinalLightCurve_WASP-12 b_2024-11-04.png (SHA-256 1fb33c9242868a80...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [406, 191], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 de13488bb2aa03c6...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 7dbc7ef

Data provenance

86 FITS frames (56 MB), WASP-12241105061515.FITS → WASP-12241105103016.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-241105.log (4a4c5196eaa7...), FinalLightCurve_WASP-12 b_2024-11-04.png (1fb33c924286...), FinalLightCurve_WASP-12 b_2024-11-04.csv (16fdf2a8f8d2...)

Compute resources

Wall time 160.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 09:29Z.